

Classical Mechanics 1 at CMI: End-semester exam, 22 Nov 2025

This question paper has 2 pages. Read questions carefully, write your answers clearly in blue and briefly explain your reasoning. Start each problem on a new sheet of paper. No computers/phones/notes/books/discussion permitted: closed book exam. 60 marks. 3 hrs.

1. **(7) Time reversal.** Suppose a particle of mass m and charge q moves in $\mathbb{R}^3$ subject to the force $q\mathbf{v} \times \mathbf{B}(\mathbf{r})$ where $\mathbf{r}(t)$ is the position of the particle and $\mathbf{v}(t) = \dot{\mathbf{r}}$. $\mathbf{B}(\mathbf{r})$ is a fixed vector field called the magnetic field.
- (a) **(1)** Write Newton's equation of motion for this particle.
 - (b) **(3)** Explain whether the motion of this particle is time-reversal invariant or not (for trajectories that exist for all time $-\infty < t < \infty$).
 - (c) **(3)** Physically interpret the equation that $\mathbf{r}(-t)$ satisfies.

New sheet

2. **(11) Anharmonic oscillator.** Consider a point particle of mass m moving along the x axis subject to an anharmonic restoring force: $f(x) = -kx - 4gx^3$ for $k, g > 0$.
- (a) **(1)** Why is the force called anharmonic?
 - (b) **(2)** Find a potential $V(x)$ corresponding to this force with the property that $V(0) = 0$. Write down the expression for the corresponding conserved energy E .
 - (c) **(3)** Find the turning points for oscillations of energy $E > 0$.
 - (d) **(5)** Find an integral expression for the time period $T(E)$ of energy- E oscillations of this particle for $E > 0$. Mention any symmetries and use them to simplify the integral expression. You are not asked to evaluate the integral.

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3. **(10) Consider a plane simple pendulum** with point-like bob of mass m suspended by a massless rod of length ℓ from a pivot subject to Earth's constant acceleration due to gravity g . Denote the counterclockwise angle of deflection from the downward vertical by θ . We orient the x axis to the right and the y axis upwards.
- (a) **(4)** Give a formula for the angular momentum component $L_z = p_\theta$. What is the phase space and coordinates on it?
 - (b) **(6)** Sketch the phase portrait of the simple pendulum. Indicate the origin and label axes. Label the static solutions and mention their stability. Draw a librational and a rotational trajectory (with arrows showing increasing time) on the phase space and label which is which.

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4. **(6) Sun-Earth system.** Given that the radius of the Sun is $r_s \approx 7 \times 10^5$ km, make order of magnitude estimates to find whether the center of mass (CM) of the Sun-Earth system and the orbit of the Sun lie within the solar interior. Recall that $1 \text{ AU} \approx 1.5 \times 10^8$ km, $m_e \approx 6 \times 10^{24}$ kg and $m_s \approx 2 \times 10^{30}$ kg. Draw a figure. Hint: The eccentricity of the orbit is very small and you are free to position the origin conveniently.

New sheet

5. (7) Recall the **Laplace-Runge-Lenz conserved vector** $\mathbf{A} = \mathbf{p} \times \mathbf{L} - m\alpha\hat{\mathbf{r}}$ that is concerned with the relative dynamics of the Kepler problem. Here, m is the reduced mass, $\alpha = Gm_1m_2$ and $\mathbf{r}$ and $\mathbf{p}$ the relative vector and relative momentum.
- (a) (4) Draw a figure of an elliptical orbit and indicate the directions of the vectors $\hat{\mathbf{r}}, \mathbf{p}, \mathbf{L}, \mathbf{A}$ appearing in this formula.
- (b) (3) We have noted that there must be relations among the conserved relative energy E , conserved relative angular momentum $\mathbf{L}$ and $\mathbf{A}$. Find one simple relation among these.

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6. (6) **Uniformly accelerated frame.** A small bob of inertial and gravitational mass m_{in} and m_{gr} is suspended by a stretched string from the roof of a car that accelerates at the rate $\mathbf{A}$. The situation is viewed from the inertial frame of the road as well as from the accelerated frame of the car. Viewed from either frame, the string is seen to settle into a configuration where it hangs at a nonzero angle relative to the vertical. If g is the magnitude of the acceleration due to gravity, show that the angle θ that the string makes with the vertical, as well as the tension T in the string are the same in both frames and find expressions for them. Is the bob displaced towards the front or back of the car? Draw force diagrams in each frame.

New sheet

7. (6) **Simultaneity.** The space-time diagram in Fig. 1 shows the world lines of two inertial observers A and B in relative motion, along with light signals to and from events E_1 and E_2 . It is drawn in a frame in which A is at rest and in units where $c = 1$. Redraw the diagram in a frame in which B is at rest and comment on the simultaneity of events E_1, E_2 for the two observers.

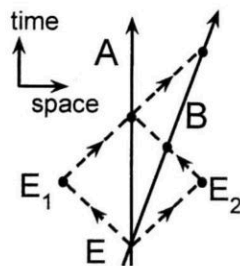


Figure 1: Space-time diagram.

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8. (7) **Light wavefront.** Suppose at time t in inertial frame S , a light wavefront lies on the sphere $x^2 + y^2 + z^2 = (ct)^2$.
- (a) (4) Show using the Lorentz boost to frame S' (moving at speed v in the x direction relative to S) that $x'^2 + y'^2 + z'^2 = (ct')^2$. We assume the frames coincide at $t = t' = 0$. Hint. Recall that $x' = \gamma(x - vt)$ and $t' = \gamma(t - vx/c^2)$. What is γ ?
- (b) (3) Interpret this result in terms of special relativity.